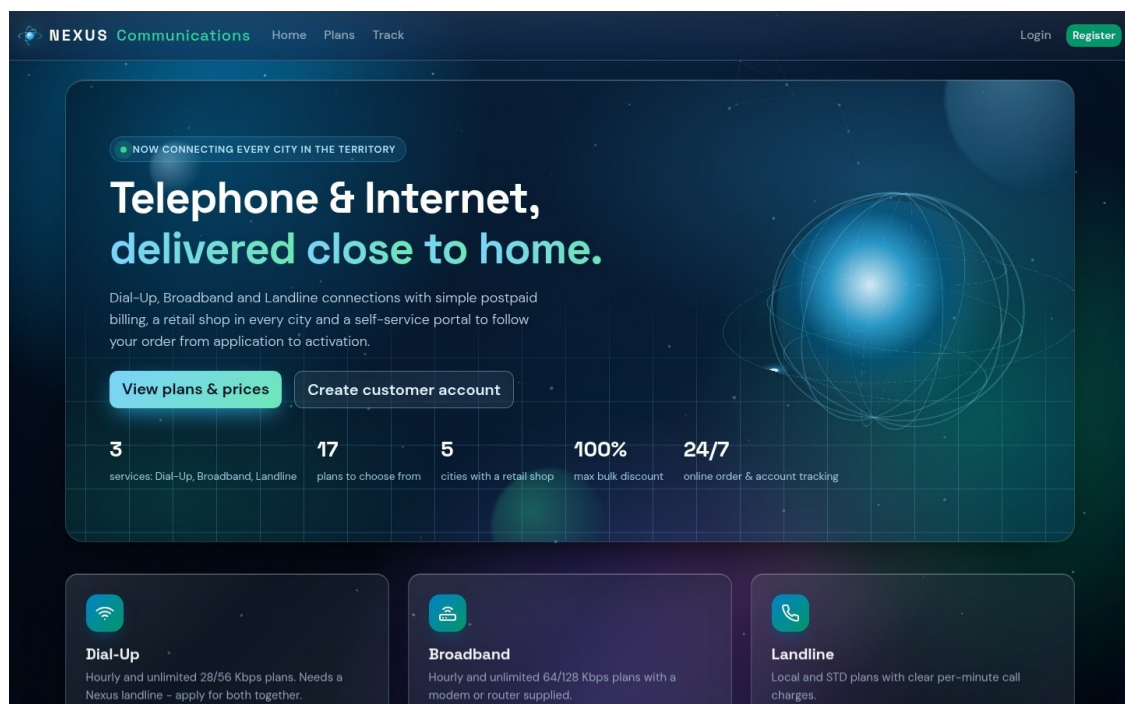


Nexus Service Marketing System

Telephone & internet service management - orders, feasibility, connections, postpaid billing and retail operations for Nexus Communications



Item	Details
Project title	Nexus Service Marketing System
Technology	ASP.NET Core 8 MVC · Entity Framework Core 8 · SQL Server · Bootstrap 5
Student name	[Your name]
Enrollment no.	[Your enrollment number]
Batch / Group	[Batch] / [Group]
Centre	[Centre name]
Submission date	[DD/MMM/YYYY]

Certificate of Completion

This is to certify that the project titled "**Nexus Service Marketing System**" has been designed, developed, tested and documented by the student(s) named on the cover page as part of the eProject learning programme, under the guidance of the faculty mentor named below.

The project fulfils the problem statement issued for the eProject: it automates the record keeping of Nexus Communications for customers, orders, feasibility checks, connections, equipment stock, vendors, retail shops, employees, plans, postpaid billing, payments and customer feedback, through separate secure logins for each role.

	Student	Faculty mentor
Name	[Your name]	[Mentor name]
Signature		
Date	[DD/MMM/YYYY]	[DD/MMM/YYYY]

Table of Contents

Certificate of Completion.....	2
Table of Contents.....	3
1. Problem Definition.....	4
1.1 Introduction.....	4
1.2 Existing scenario.....	4
1.3 Proposed solution.....	4
1.4 Objectives.....	4
2. Customer Requirement Specification.....	6
3. Project Plan.....	8
4. System Design.....	9
4.1 Architecture.....	9
4.2 Roles and functions.....	10
4.3 Entity-Relationship Diagram.....	11
4.4 Data dictionary.....	11
4.5 Identifier formats.....	12
5. Algorithms.....	13
5.1 Order placement.....	13
5.2 Feasibility check.....	13
5.3 Connection creation.....	13
5.4 Bill generation.....	14
5.5 Payments and overdue connections.....	14
6. GUI Standards.....	15
6.1 Colour palette.....	15
6.2 Typography and layout.....	15
6.3 Motion and accessibility.....	15
6.4 Glass interaction states.....	16
7. Interface Design Document.....	17
8. Login Credentials and How to Run.....	26
8.1 Requirements.....	26
8.2 Steps.....	26
8.3 Login credentials.....	26
9. Task Sheet.....	27
10. Project Review and Monitoring Report.....	28
11. Unit Testing Checklist.....	29
12. Final Checklist.....	31
Appendix - Plan price list.....	31

Right-click the table and choose Update Field (or press F9) to refresh page numbers after editing.

1. Problem Definition

1.1 Introduction

Nexus Communication is an organisation that provides telecommunication and internet services. It offers two types of internet connection - **Dial-Up** (which needs a Nexus landline) and **Broadband** - as well as **Landline** telephone connections. To meet growing demand it has opened retail shops in every city of its territory, where customers can enquire, place orders for new connections and pay their bills.

Before a connection is given, the company checks feasibility: the distance from the exchange and the availability of server capacity. Equipment such as modems and routers is bought from different manufacturing vendors, so stock must be tracked against the orders received.

1.2 Existing scenario

- Customer, order, connection, payment and vendor records are kept on paper, so slip-ups occur and searching is slow.
- Bulk and corporate schemes give 25% to 100% discount on the advance and security deposit depending on the number of connections, and are calculated by hand.
- Address proofs, application forms and receipts are filed manually by year and area.
- Bills, dues, replacement charges and discounts are worked out manually for every customer.

1.3 Proposed solution

An online application with a single SQL Server database and a separate login for each role - Admin, Accounts, Technical, Retail employee and Customer - that updates and maintains every record automatically:

- unique 11-character **order numbers** (D / B / T + 10-digit serial) and 16-character **account IDs** (type + 3-digit city code + 12-digit serial), generated by the system;
- search by order number or account ID, plus an **advanced search** by name, contact number, connection type and date or period;
- postpaid **billing** by the Accounts department with plan charges, call charges, deposits, discounts and **12.24% service tax**, with paid and due amounts always visible;
- **feasibility** checks, connection creation and activation status (active, temporarily inactive, permanently inactive) handled by Technical staff;
- customer self-service for contact details, orders, bills, documents and feedback.

1.4 Objectives

#	Objective	Achieved by
1	Replace paper records with one database	19 related tables in SQL Server, created by EF Core migrations
2	Give each role only the functions it needs	Cookie login, hashed passwords, role-based areas
3	Generate order numbers and account IDs automatically	IdentifierGenerator with atomic counters and CHECK constraints

#	Objective	Achieved by
4	Calculate bills exactly as the specification says	BillingService (plan validity, call rates, deposit, discount, tax)
5	Let anyone track an order or connection	Public Track page; amounts shown only to authorised users
6	Professional, responsive and accessible interface	Custom design system with motion, keyboard and reduced-motion support

2. Customer Requirement Specification

Every requirement of the problem statement, where it is implemented, and its status.

No.	Requirement (problem statement)	Implementation	Status
1	Separate logins for Admin, Accounts, Technical, Retail employees and Customers	Role areas, cookie authentication, hashed passwords	Done
2	Admin maintains employees, stock, vendors, retail shops and plans (insert, update, delete, search)	Admin area: Cities, Shops, Employees, Plans, Discounts, Vendors, Products, Purchases	Done
3	Order ID: 11 characters, D / T / B + serial	IdentifierGenerator + database CHECK constraint	Done
4	Account ID: type + 3-digit city code + 12-digit serial (16 characters)	IdentifierGenerator, Cities.Code	Done
5	Orders placed by retail employees (and by customers themselves)	Retail > Orders, Customer > Orders	Done
6	Feasibility: landline + internet when both applied for; internet only when the customer already has a Nexus landline	Dial-up order with landline plan; OrderWorkflowService	Done
7	Technical: update order status, create connections, temporary / permanent inactive, maintain equipment	Technical area: Feasibility, Connections, Overdue, Equipment	Done
8	Bills generated and updated only by Accounts; status of the connection depends on the bill	Accounts > Bills; Technical > Overdue suspends unpaid connections	Done
9	Payments by retail outlets and the Accounts office; paid and due amounts shown	Retail > Collect payment, Accounts > Record payment	Done
10	Bulk discount 25% / 50% / 75% / 100% on advance and security deposit	DiscountSchemes, counted over all the customer's connections	Done
11	Security deposits 325 / 500 / 250 \$ and the full plan price list	Plans table (17 active plans + yearly STD awaiting price)	Done
12	Service tax 12.24% on the whole bill	BillingService	Done
13	Search order status by order ID and connection status by account ID	Public Track page	Done
14	Advanced search: ID, name, connection type, date or period, contact number	Admin > Advanced search, staff Search page	Done
15	Customer: contact details, place and track orders, check bills	Customer portal	Done

No.	Requirement (problem statement)	Implementation	Status
16	Address proofs, application forms and receipts filed by year and area	Customer Documents (PDF / PNG / JPEG, filed by year and city)	Done
17	Billing considers discounts, schemes and payments made previously	First-bill discount, balance brought forward on every bill	Done
18	Customer feedback collected	Customer > Feedback, Admin > Feedback	Done

3. Project Plan

The project followed seven phases. Each phase ended with a working, testable result.

Phase	Work	Output	Week
1. Analysis	Study the problem statement, roles and business rules	Requirement list (chapter 2)	1
2. Design	ER diagram, identifier formats, screen list, GUI standards	Database design, interface plan	1-2
3. Database	EF Core entities, configurations, CHECK constraints, seed data	19 tables, 5 migrations	2
4. Development	Five role areas and the business services	Working web application	2-3
5. Interface	Design system, 3D hero, motion, dashboards, responsive layout	site.css, site.js, views	3
6. Testing	Unit testing checklist, 48 automated end-to-end checks	Test results (chapter 11)	3-4
7. Delivery	Documentation, presentation, database backup, source ZIP	Final submission	4

4. System Design

4.1 Architecture

The application is a layered ASP.NET Core 8 MVC website. Each role has its own area protected by role-based authorization. Business rules live in services, not in controllers, so every screen applies the same rules. Entity Framework Core maps the entities to SQL Server.

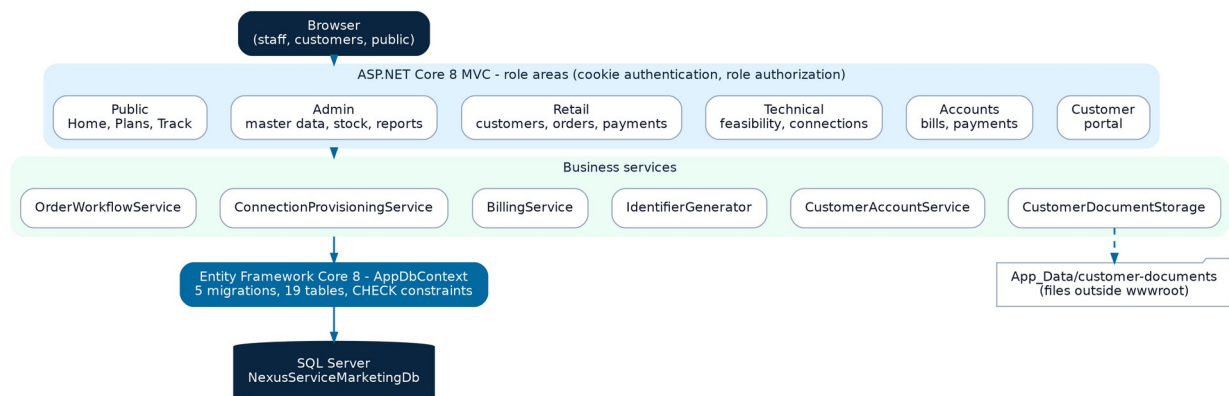


Figure 1 - System architecture

4.2 Roles and functions

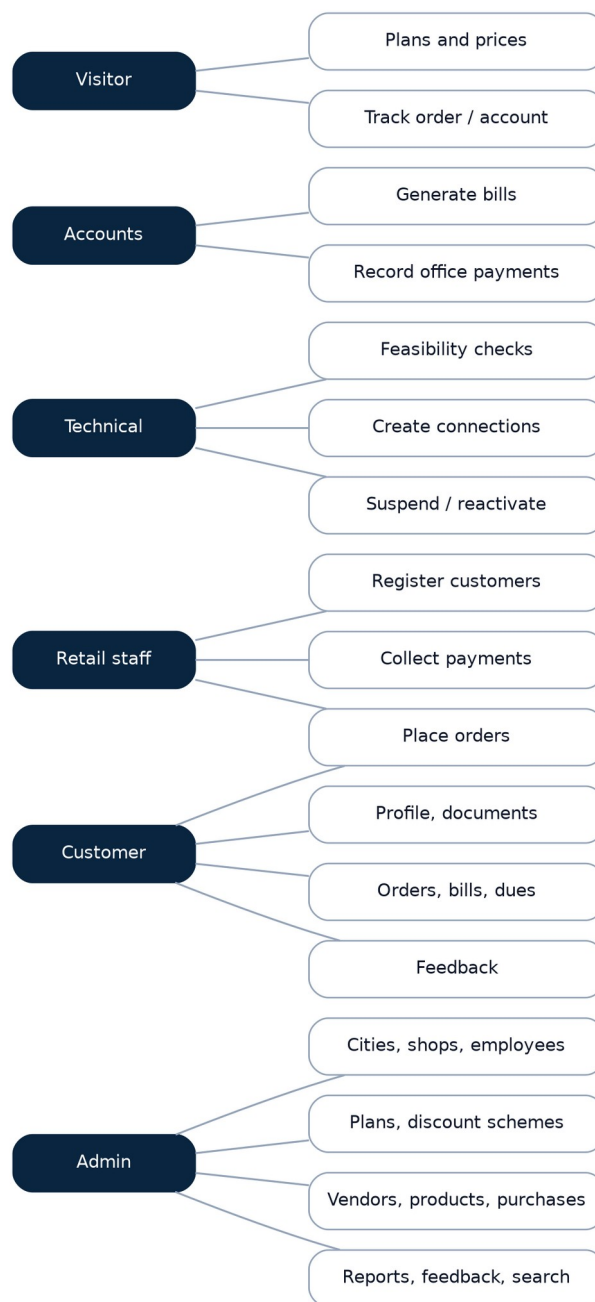


Figure 2 - What each role can do

4.3 Entity-Relationship Diagram

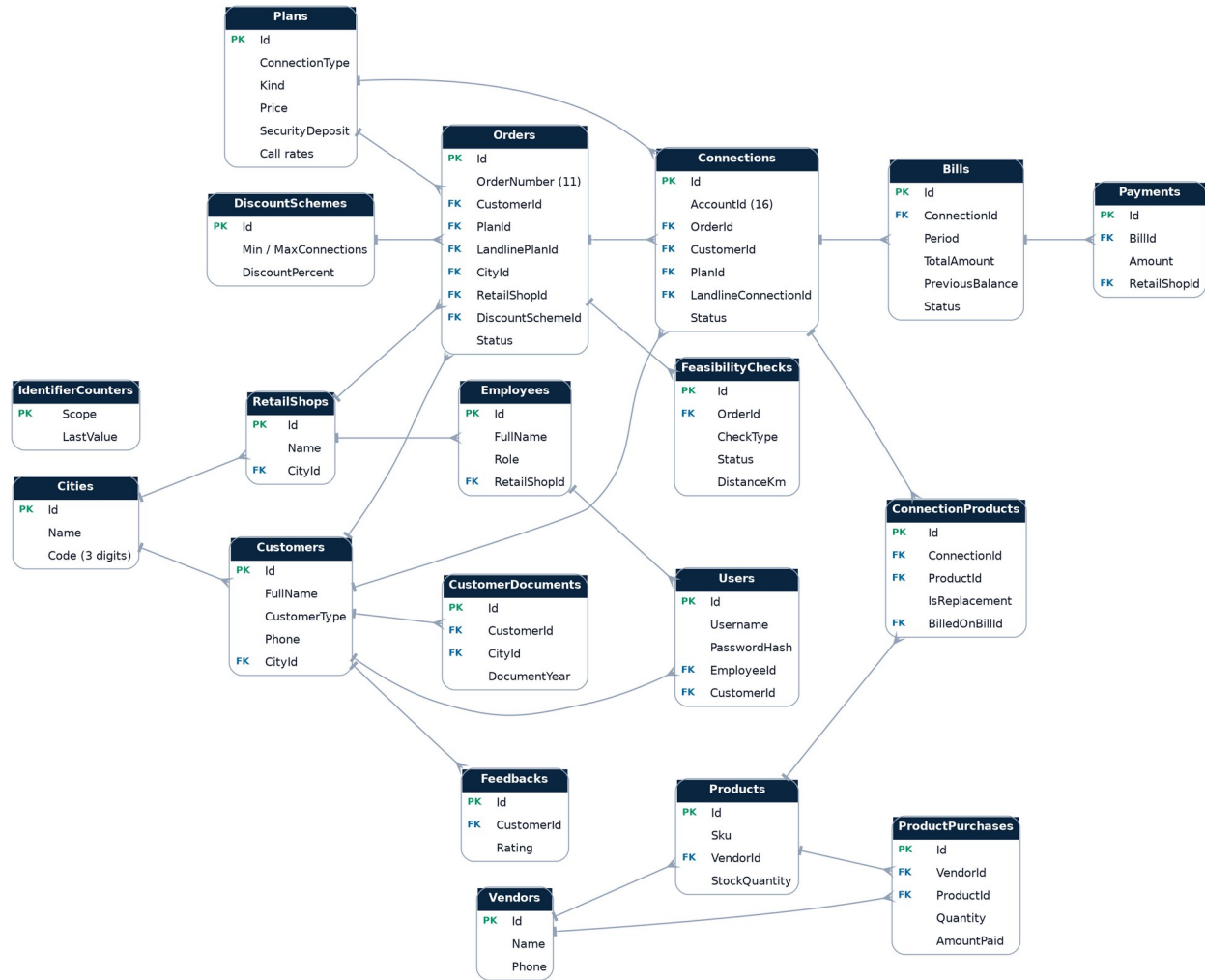


Figure 3 - Entity-relationship diagram (PK primary key, FK foreign key; crow's foot = many)

4.4 Data dictionary

Table	Purpose	Key rules
Users	Login accounts for staff and customers	Unique username, hashed password
Employees	Admin, accounts, technical and retail staff	Retail staff must belong to a shop
Customers	Personal and contact details	Individual or corporate
Cities	Cities of the territory	Unique 3-digit code used in account IDs
RetailShops	Retail outlets in each city	Unique name per city
Plans	Plan catalogue and charges	Hourly / unlimited / local / STD rules per kind
DiscountSchemes	Bulk discount bands	No overlapping bands at order time

Table	Purpose	Key rules
Orders	Applications for connections	Order number format D/B/T + 10 digits
FeasibilityChecks	Landline / internet feasibility results	Distance and server capacity
Connections	Connections provided	Account ID format; status values
ConnectionProducts	Equipment issued and replaced	Replacement charge billed once
Vendors, Products	Suppliers and equipment stock	Stock never negative
ProductPurchases	Purchases from vendors and amounts paid	Purchase increases stock
Bills	Postpaid bills	One bill per connection and period
Payments	Payments received	Never more than the amount due
Feedbacks	Customer feedback	Rating 1-5
CustomerDocuments	Filed proofs, forms and receipts	Filed by year and city
IdentifierCounters	Serial counters for IDs	Atomic increment under concurrency

4.5 Identifier formats

Identifier	Format	Example
Order number	Type letter (D / B / T) + 10-digit serial = 11 characters	D0000000001
Account ID	Type letter + 3-digit city code + 12-digit serial = 16 characters	B001000000000001

5. Algorithms

5.1 Order placement

1. Retail employee (or the customer) chooses customer, service type, plan, city, address and quantity.
2. The plan must be active and of the chosen service type; the city must be active.
3. Dial-up: if the customer has no active Nexus landline a **landline plan is required** - the telephone line and dial-up are applied for together.
4. Bulk band: count the customer's connections (not permanently closed) + open orders + this order; attach the matching discount scheme.
5. Take the next serial for the type atomically and build the order number; status = Placed.

5.2 Feasibility check

1. Technical starts the checks: Telephone - landline check; Broadband - internet check; Dial-up - both checks when a line is applied for, otherwise internet only.
2. For each check record the distance and, for internet, server capacity.
3. Any check not feasible - order Not feasible. All feasible - order Feasible.

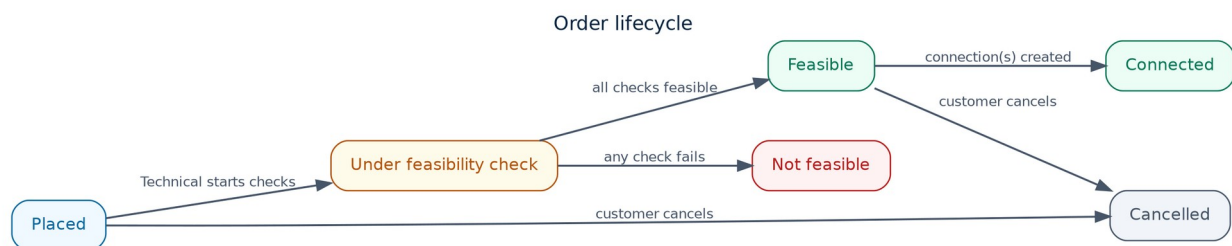


Figure 4 - Order lifecycle

5.3 Connection creation

1. Order must be Feasible. Internet orders take one modem / router per connection from stock.
2. Deposit = plan deposit, reduced by the bulk discount when the scheme covers deposits.
3. For a bundled dial-up order, first create each telephone line (with its own number), then the dial-up connection linked to it.
4. Account ID = type + city code + next 12-digit serial for that type and city. Status Active; order Connected.

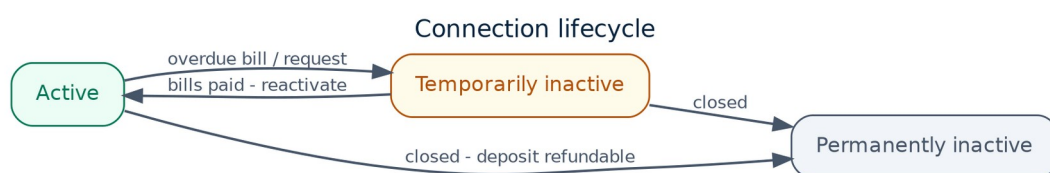


Figure 5 - Connection lifecycle

5.4 Bill generation

1. Plan fee: charged only when the period starts after the date the current plan is already paid to (plan validity).
2. Call charges = local x local rate + STD x STD rate + mobile x mobile rate (landline plans only).
3. Add unbilled equipment replacement charges and mark them billed.
4. First bill only: add the security deposit and subtract the bulk discount on the advance.
5. SubTotal = plan fee + usage + deposit + replacements - discount; tax = 12.24% of SubTotal; Total = SubTotal + tax.
6. Record the unpaid balance of earlier bills as **brought forward**.

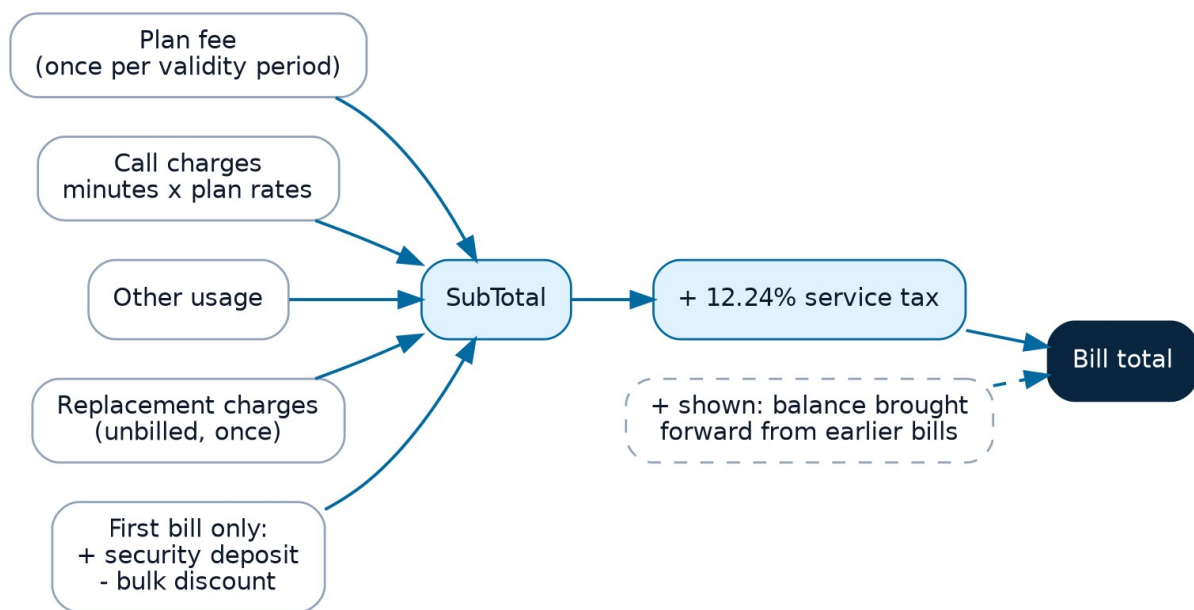


Figure 6 - How a bill is built

Worked example (verified by the automated tests)

Landline Local Monthly plan, first bill, 100 local minutes at \$0.75:

Plan fee \$35.00 + calls \$75.00 + deposit \$250.00 = SubTotal \$360.00

Service tax 12.24% = \$44.06 · Bill total = **\$404.06**

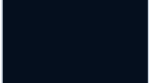
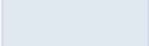
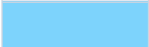
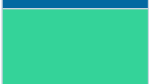
5.5 Payments and overdue connections

1. A payment can never exceed what is still due; the bill becomes Partially paid or Paid.
2. Retail staff collect only bills of orders handled by their shop or self-service orders.
3. An active connection with a bill past its due date and not fully paid appears in Technical > Overdue and can be set temporarily inactive; once every bill is paid it is listed for reactivation.

6. GUI Standards

The interface follows one design system, defined as tokens in `wwwroot/css/site.css`, so every screen looks and behaves the same. It uses a dark "glass" theme: every page sits on an animated background and the work surfaces are frosted dark glass.

6.1 Colour palette

Swatch	Role	Hex	Use
	Background	#050F1E	Deep navy page canvas behind the animated aurora
	Glass surface	rgba(13,28,50,.62)	Cards, tables, forms - frosted glass (22 px blur); turns clear on hover
	Text	#E2E8F0	Body text (12:1 contrast on the glass)
	Link / highlight	#7DD3FC	Links, table headers, icons
	Primary	#0369A1	Buttons, active states (white text 5.9:1 contrast)
	Accent	#34D399 / #059669	Success, call to action
	Danger	#F87171	Errors, overdue

6.2 Typography and layout

Element	Standard
Headings	Space Grotesk, 600-700 weight (bundled with the app)
Body text	DM Sans 16 px, line height 1.55
Grid	Bootstrap 5 responsive grid; tested at 390, 768, 1024 and 1440 px
Navigation	Sticky glass bar, role menu, account drop-down; active page highlighted
Page header	Title bar with breadcrumb on inner pages; 3D stage on home, login and dashboards
Forms	Labels above fields, validation next to the field, forms shown as cards, submit button shows a spinner
Tables	Glass background, uppercase headers, row hover, coloured status badges, scroll sideways on phones
Money	US dollars, two decimals

6.3 Motion and accessibility

- Animated page background: slowly drifting aurora glows, a moving grid and a particle network that reacts to the mouse.
- Loader on the first page of a visit; page cross-fade with a top progress bar.
- 3D globe and parallax layers on the hero, tilt and glare on cards, magnetic and ripple buttons.

- Clear-glass interaction: clickable cards are frosted at rest; on hover or keyboard focus the frost clears, the rim lights up and the card lifts; on click it presses in and a soft light spreads from the click point.
- Scroll reveal and count-up figures; all motion uses transform and opacity only.
- Every animation is switched off for users who choose reduced motion; hover effects are skipped on touch screens.
- Visible keyboard focus, skip-to-content link, text contrast of at least 4.5:1, labels on every field.

6.4 Glass interaction states

State	What the user sees	How it is built
Rest	Frosted glass card, faint white rim, soft shadow	Translucent fill + backdrop-filter blur(22px) saturate(170%)
Hover / focus	Frost clears so the background shows through; bright blue rim and glow; card lifts 4 px	Blur eases to 6 px in 0.45 s; inset rim + outer glow; translateY(-4px)
Press (click)	Card sinks slightly; a light flash spreads from the click point	scale(.985) on :active; site.js adds a radial flash that fades in 0.7 s
Current page	Active menu item shown as a small clear-glass button	Glass gradient + inset highlight on .nav-link.is-active
Table row	Light glass highlight with a blue left edge	Gradient background + inset 2 px accent line
No blur support	Solid dark panels, text stays readable	@supports fallback to an opaque surface

7. Interface Design Document

Screens of the finished application, taken from the running system with sample data.

Role	Main screens
Public	Home, Plans and prices, Track order / account, Login, Register
Admin	Dashboard, Cities, Retail shops, Employees, Plans, Discount schemes, Vendors, Products, Purchases, Customers, Orders, Advanced search, Reports, Feedback
Retail	Dashboard, Customers, Place order, Orders, Connections, Collect payment, Payment history, Search
Technical	Dashboard, Feasibility, Connections (create, status, plan, replace equipment), Overdue, Equipment, Search
Accounts	Dashboard, Generate bill, Bills and payments, Search
Customer	Dashboard, Profile, Orders (place, track, cancel), Connections, Bills, Documents, Feedback

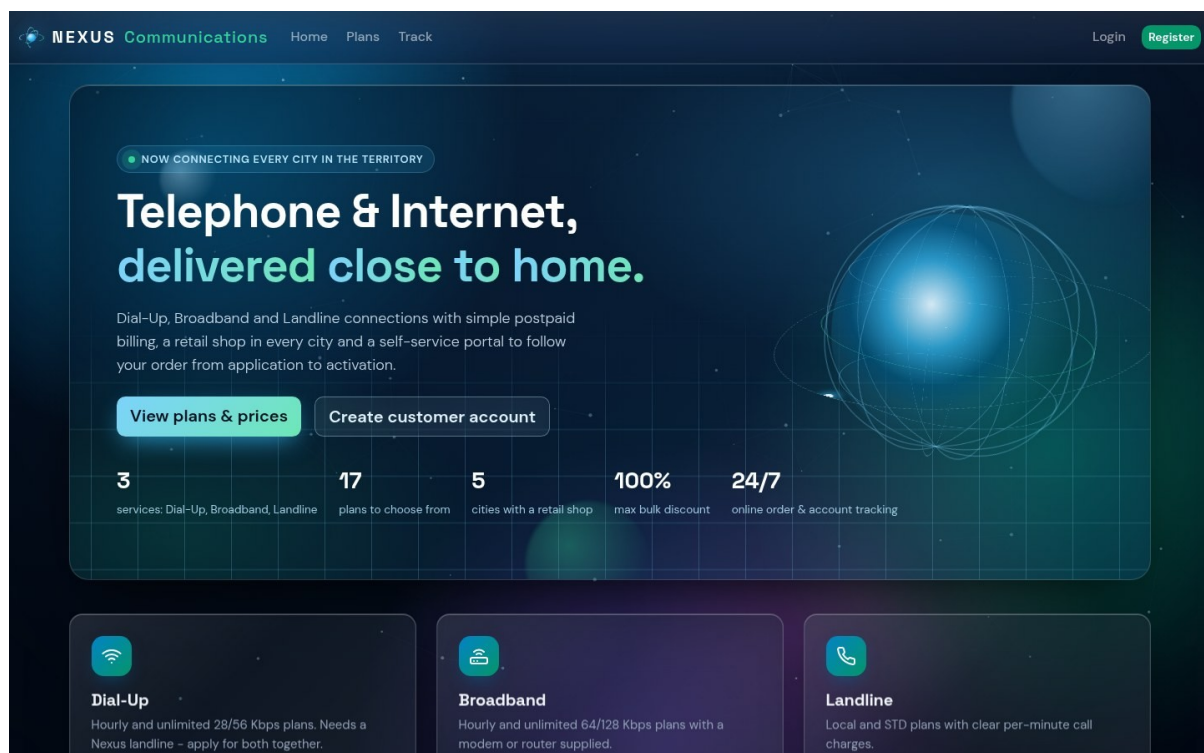


Figure 7 - Home page - 3D parallax hero, live figures and services

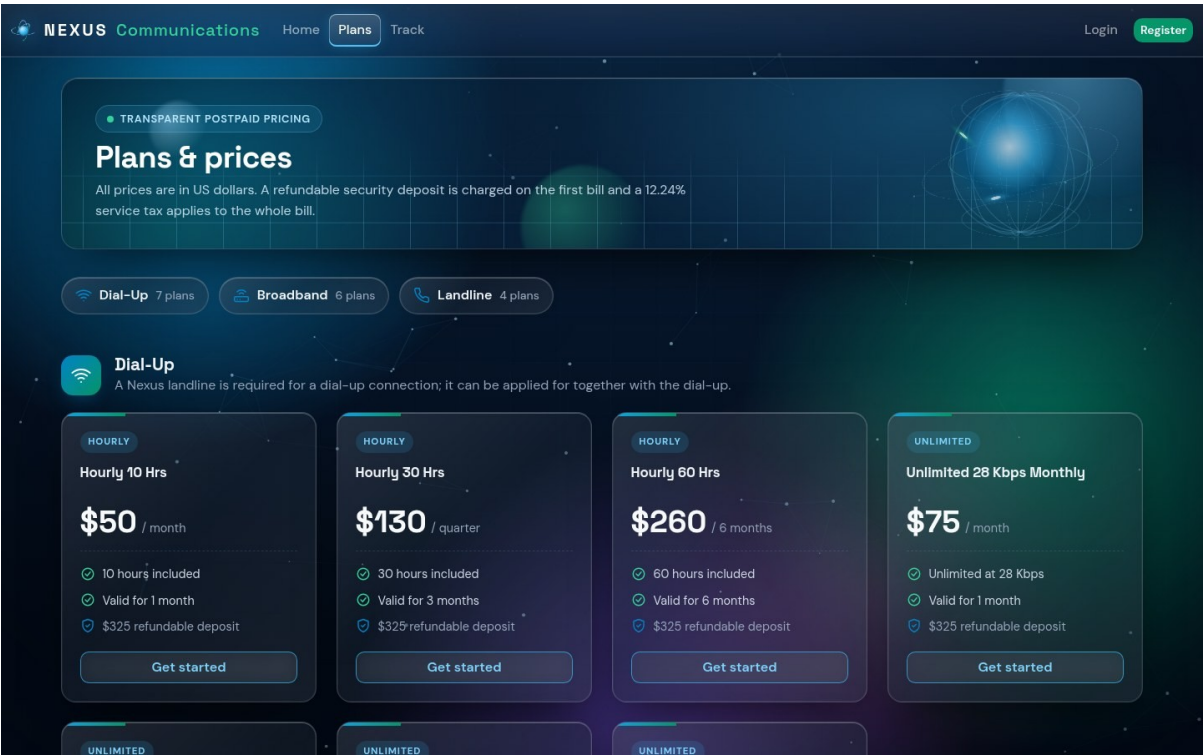


Figure 8 - Plans and prices - pricing cards grouped by service

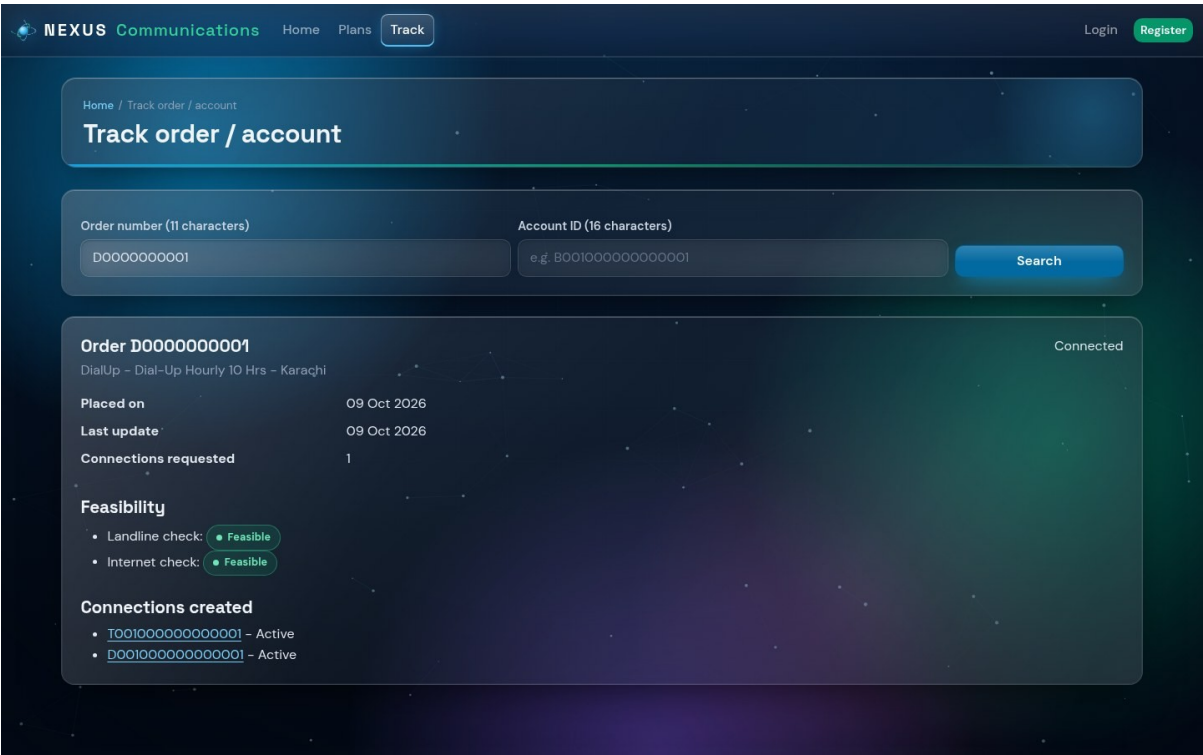


Figure 9 - Public tracking of an order number

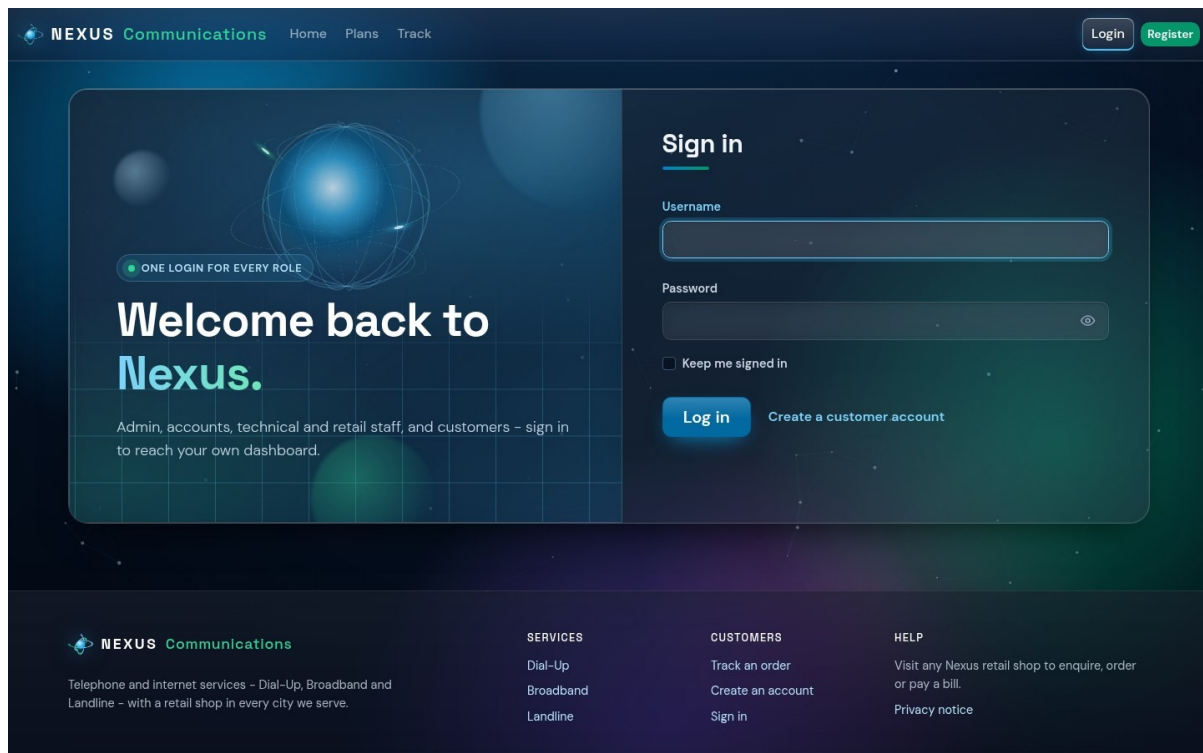


Figure 10 - Sign in - one login page for every role

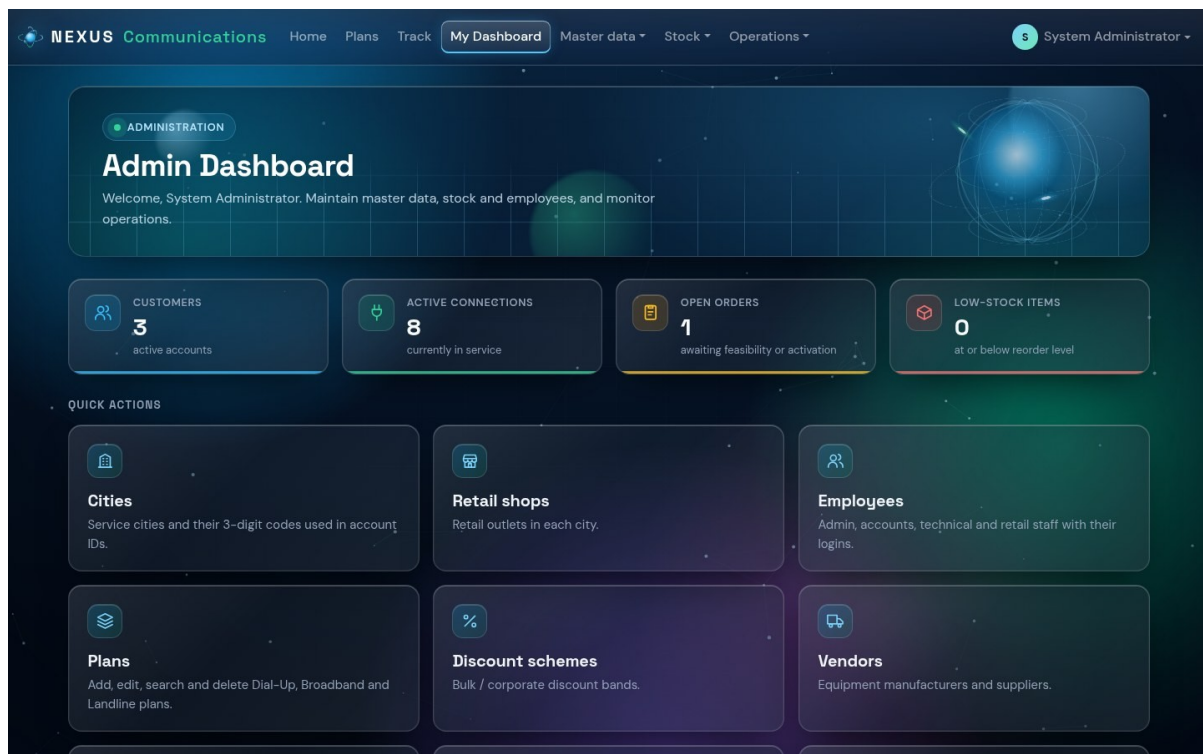


Figure 11 - Admin dashboard - live key figures and quick actions

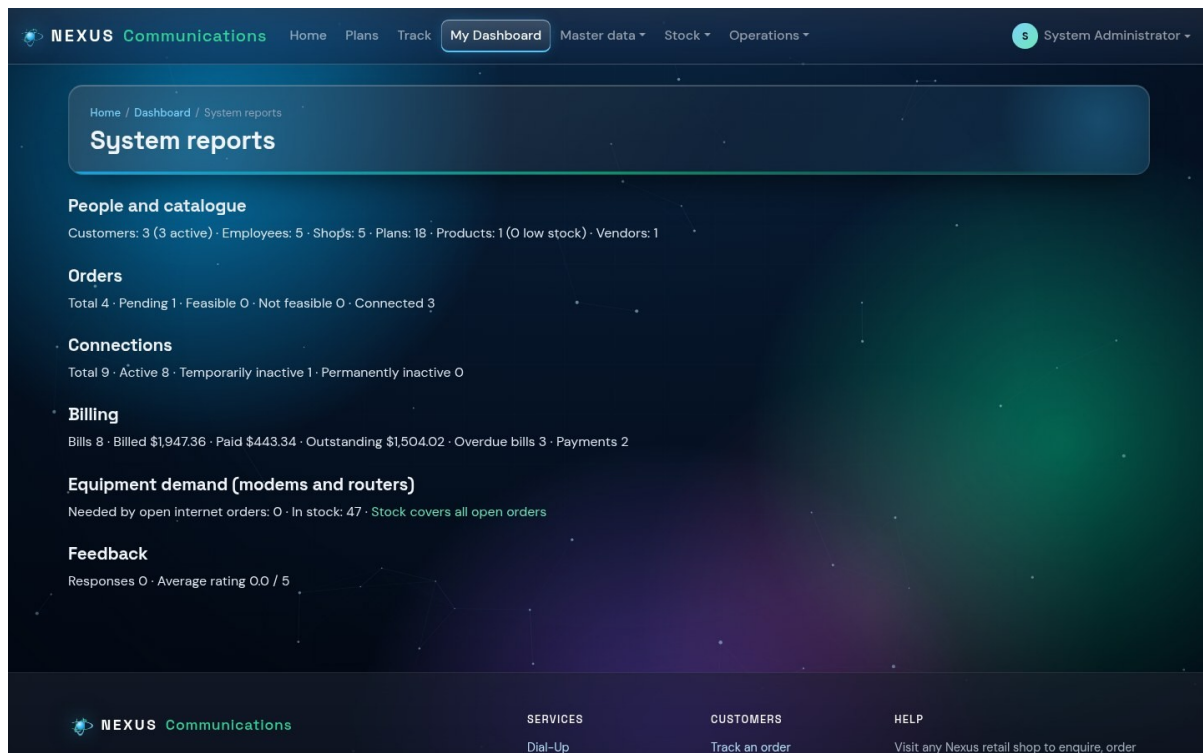


Figure 12 - Admin reports - orders, connections, billing and equipment demand

The screenshot displays the 'Place a customer order' form in the Nexus Communications Retail interface. The header shows the logo and navigation links: Home, Plans, Track, My Dashboard, Customers, Orders, Connections, Collect payment, Payment history, Search, and a retail link. The form is titled 'Place a customer order' and contains the following fields:

- Customer:** Choose customer (dropdown)
- Service type:** DialUp (dropdown)
- Plan:** Choose plan (dropdown)
- Landline plan (dial-up without a Nexus landline):** Customer already has a Nexus landline (dropdown)
- Installation city:** Choose city (dropdown)
- Installation address:** Text input field
- Number of connections:** 1 (input field)

A note below the landline plan dropdown states: 'Dial-up needs a Nexus landline. If there is none, choose a landline plan: the telephone line and the dial-up are applied for together (both feasibility checks are run).'

Figure 13 - Retail - placing an order (dial-up with a new landline)

Home / Dashboard / Collect customer payments

Collect customer payments

CUSTOMER	ACCOUNT	BILL	OUTSTANDING	STATUS	
Bob Broad	BO01000000000001	#3	1010.16	Overdue	Collect payment
Bob Broad	BO01000000000001	#6	448.96	Unpaid	Collect payment
Bob Broad	BO01000000000001	#7	44.90	Unpaid	Collect payment

NEXUS Communications
Telephone and internet services – Dial-Up, Broadband and Landline – with a retail shop in every city we serve.

SERVICES
Dial-Up
Broadband
Landline

CUSTOMERS
Track an order
Create an account
Sign in

HELP
Visit any Nexus retail shop to enquire, order or pay a bill.
[Privacy notice](#)

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Postpaid billing : 12.24% service tax included on every bill

Figure 14 - Retail - bills the shop can collect

Home / Dashboard / Orders and feasibility

Orders and feasibility

ORDER	CUSTOMER	SERVICE	STATUS	CHECKS	ACTIONS
T0000000001	Corp Ltd	Telephone	Placed		Start checks
T0000000002	Corp Ltd	Telephone	Connected	Landline: Feasible	
B0000000001	Bob Broad	Broadband	Connected	Internet: Feasible	
D0000000001	Alice Dial	Dial-Up	Connected	Landline: Feasible Internet: Feasible	

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Postpaid billing : 12.24% service tax included on every bill

Figure 15 - Technical - feasibility queue

NEXUS Communications Home Plans Track My Dashboard Feasibility Connections **Overdue** Equipment Search

Home / Dashboard / Overdue bills

Overdue bills

Connections are postpaid: an active connection whose bill is past its due date and not fully paid is set **temporarily inactive**. When the customer has paid, reactivate it from the second table.

Active connections with overdue bills (0)

ACCOUNT	CUSTOMER	PHONE	OVERDUE BILLS	OVERDUE AMOUNT	OLDEST DUE DATE
No active connection has an overdue bill.					

Suspended connections with all bills paid (0)

ACCOUNT	CUSTOMER	PHONE
None.		

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Landline

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Privacy notice

Figure 16 - Technical - overdue bills: suspend and reactivate

NEXUS Communications Home Plans Track **My Dashboard** Bills & payments Search

ACCOUNTS DEPARTMENT

Accounts Dashboard

Welcome, acct1. Generate bills (12.24% service tax) and record payments received at the office.

OUTSTANDING
\$1,504
unpaid on open bills

OVERDUE BILLS
3
past their due date

COLLECTED THIS MONTH
\$443
payments received

BILLS ISSUED
8
all time

QUICK ACTIONS

Bills & payments
See all bills, outstanding balances and record payments.

Generate a bill
Issue a new postpaid bill for a connection.

Plans & prices
Reference of plan charges and security deposits.

NEXUS Communications
Telephone and internet services – Dial-Up, Broadband and Landline – with a retail shop in every city we serve.

SERVICES
Dial-Up
Broadband
Landline

CUSTOMERS
Track an order
Create an account
Sign in

HELP
Visit any Nexus retail shop to enquire, order or pay a bill.
Privacy notice

Figure 17 - Accounts dashboard

Generate bill

Charges are calculated by the system: the plan fee (once per plan validity period), call charges from the minutes entered and the plan's per-minute rates, unbilled equipment replacements, the security deposit and bulk discount on the first bill, and 12.24 % service tax on the whole bill.

Connection: T001000000000001 - Alice Dial

Billing period starts: 12/01/2026

Show charges

T001000000000001 - Landline Local Monthly

- Plan fee on this bill: **\$35.00**
- Local calls: \$0.75 / minute
- Equipment replacement charges to bill: \$0.00
- Unpaid from earlier bills (brought forward): \$0.00

Billing period starts: 12/01/2026

Billing period ends: 12/31/2026

Issue date: 10/09/2026

Due date: 10/23/2026

Local call minutes: 0

Other usage charges (e.g. extra internet hours):

Figure 18 - Accounts - generating a bill with the charges preview

Bills

Generate bill

"Brought forward" is what was still unpaid on the connection's earlier bills when the bill was issued; it is paid against those bills.

BILL	CUSTOMER	ACCOUNT	PERIOD	TOTAL	BROUGHT FORWARD	PAID	OUTSTANDING	STATUS
#8	Bob Broad	B001000000000001	11/1/2026 - 11/30/2026	\$0.00	\$1,504.02	\$0.00	\$0.00	Paid
#2	Alice Dial	T001000000000001	10/1/2026 - 10/31/2026	\$39.28	\$404.06	\$39.28	\$0.00	Paid
#7	Bob Broad	B001000000000001	10/1/2026 - 10/31/2026	\$44.90	\$1,459.12	\$0.00	\$44.90	Unpaid Record payment
#1	Alice Dial	T001000000000001	9/1/2026 - 9/30/2026	\$404.06	\$0.00	\$404.06	\$0.00	Paid
#6	Bob Broad	B001000000000001	9/1/2026 - 9/30/2026	\$448.96	\$1,010.16	\$0.00	\$448.96	Unpaid Record payment
#5	Bob Broad	B001000000000001	8/1/2026 - 8/31/2026	\$0.00	\$1,010.16	\$0.00	\$0.00	Paid
#4	Bob Broad	B001000000000001	7/1/2026 - 7/31/2026	\$0.00	\$1,010.16	\$0.00	\$0.00	Paid

Figure 19 - Accounts - all bills with status badges

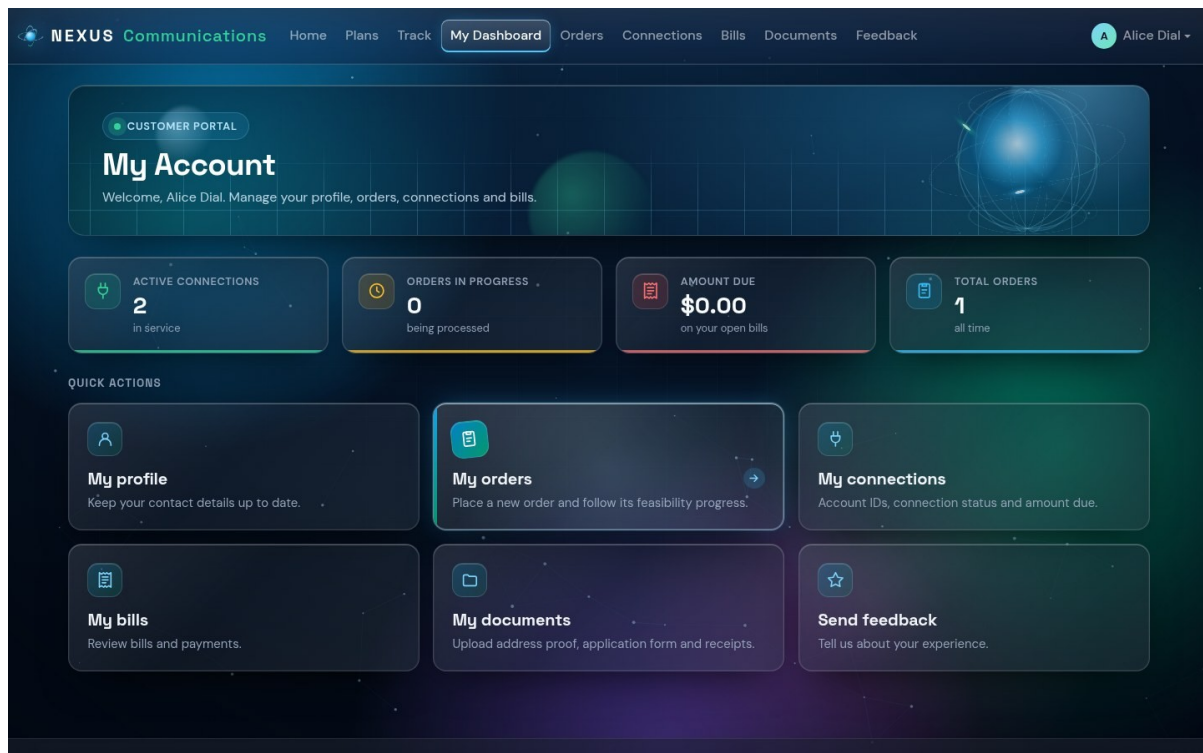


Figure 20 - Customer portal dashboard

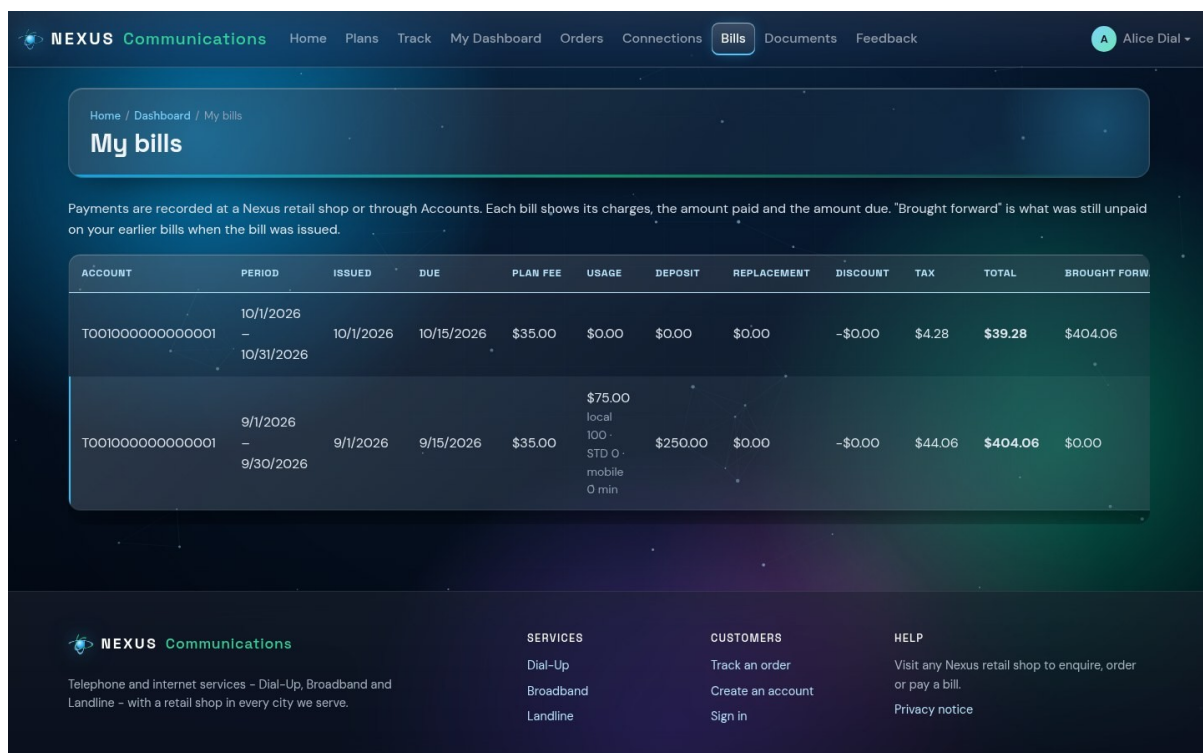


Figure 21 - Customer - itemised bills and balance

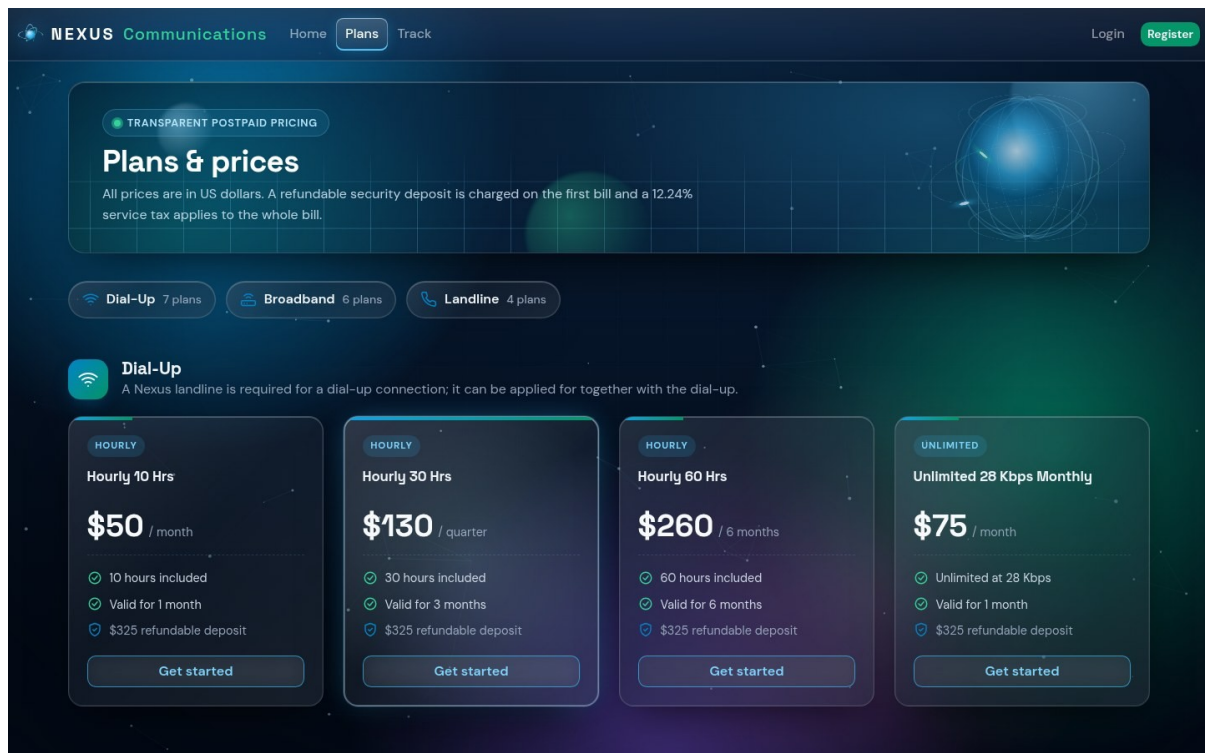


Figure 22 - Clear-glass hover - the hovered plan card turns clear with a lit rim

8. Login Credentials and How to Run

8.1 Requirements

- .NET 8 SDK and SQL Server (Express, LocalDB, Developer or Standard).
- Connection string in appsettings.json (default: .\SQLEXPRESS, Windows login).

8.2 Steps

1. Open a terminal in the folder that contains NexusServiceMarketingSystem.csproj.
2. dotnet tool install --global dotnet-ef --version 8.*
3. dotnet ef database update (or run Database\NexusServiceMarketingDb-schema.sql in SSMS)
4. dotnet run, then open the address shown (http://localhost:5294).

8.3 Login credentials

Role	Username	Password	Notes
Admin	admin	Admin@123	Created automatically; change after first login
Accounts / Technical / Retail	[as created]	[as created]	Created by Admin > Employees
Customer	[as registered]	[as registered]	Self-registration or created by staff
Database	Windows login / sa	[your SQL password]	NexusServiceMarketingDb

9. Task Sheet

#	Task	Module	Assigned to	Status
1	Requirement analysis and use cases	Analysis	[Member]	Completed
2	ER diagram and database design	Design	[Member]	Completed
3	Entities, configurations and migrations	Database	[Member]	Completed
4	Login, roles and password hashing	Security	[Member]	Completed
5	Admin master data and stock	Admin	[Member]	Completed
6	Orders, bulk discount, bundled landline	Retail / Customer	[Member]	Completed
7	Feasibility and connection provisioning	Technical	[Member]	Completed
8	Billing, payments, overdue suspension	Accounts	[Member]	Completed
9	Tracking and advanced search	Public / Staff	[Member]	Completed
10	Customer documents and feedback	Customer	[Member]	Completed
11	Design system, 3D hero, dashboards	Interface	[Member]	Completed
12	Testing and documentation	Quality	[Member]	Completed

10. Project Review and Monitoring Report

Review point	Finding	Action taken
Combined telephone + dial-up applications	Orders could pass both checks but never be connected	Landline plan on the order; provisioning creates the line and links the dial-up
Billing completeness	Old dues and plan validity were not considered	Balance brought forward; plan fee charged once per validity period
Charges calculation	Usage was typed in by hand	Call charges calculated from minutes and plan rates
Replacement charges	Could be forgotten on the next bill	Recorded with the replacement and billed automatically once
Postpaid status	Unpaid bills did not affect connections	Technical > Overdue: suspend and reactivate
Document upload	Stored files lost their first bytes	Fixed; downloads are byte-identical
Coding standard	Some controllers had no comments	Every controller and service commented
Interface	Plain template look	Design system, 3D hero, dashboards with live figures

11. Unit Testing Checklist

The checks below were run automatically against the running application on SQL Server 2022 (48 end-to-end checks plus browser checks). All passed.

#	Module	Test case	Expected result	Result
1	Orders	Dial-up order without a landline plan	Rejected with a clear message	Pass
2	Orders	Combined telephone + dial-up order	Landline plan stored; both feasibility checks created	Pass
3	Orders	Bulk order: second order of 6 (12 connections)	25% bulk band applied	Pass
4	Feasibility and connections	All checks feasible	Order becomes Feasible	Pass
5	Feasibility and connections	Provision combined order without a phone number	Rejected - number required	Pass
6	Feasibility and connections	Provision combined order	Telephone line + dial-up created and linked; 16-character account IDs	Pass
7	Feasibility and connections	Deposits	Landline \$250, Dial-up \$325; 25% band gives \$187.50	Pass
8	Feasibility and connections	6 telephone lines with one number	Rejected - one number per line	Pass
9	Billing	Bill preview	Shows plan rate, replacement charges and brought-forward balance	Pass
10	Billing	First landline bill, 100 local minutes	$35 + 75 + 250 = 360$; tax 44.06; total 404.06	Pass
11	Billing	Call minutes on an internet connection	Rejected	Pass
12	Billing	STD minutes on a local-only plan	Rejected	Pass
13	Billing	Second monthly bill	Monthly fee charged again; 404.06 brought forward	Pass
14	Billing	Quarterly plan billed monthly	Plan fee only in June and September	Pass
15	Billing	Equipment replacement	40.00 charged on the next bill only, then marked billed	Pass
16	Payments and status	Overdue page	Lists only active connections with overdue bills	Pass
17	Payments and status	Suspend all	Overdue connections temporarily inactive; others unchanged	Pass

#	Module	Test case	Expected result	Result
18	Payments and status	Retail collects all bills	Bills Paid; connection offered for reactivation	Pass
19	Payments and status	Reactivate	Connection Active	Pass
20	Payments and status	Other shop opens a bill's payment form	Not found (shop isolation)	Pass
21	Documents, pages and interface	Upload and download a PDF	Byte-identical file returned	Pass
22	Documents, pages and interface	All role pages	Every page renders without errors	Pass
23	Documents, pages and interface	Phone width 390 px	No horizontal scrolling	Pass
24	Documents, pages and interface	Reduced motion	No loader; all content visible immediately	Pass

12. Final Checklist

Item	Status
All requirements of the problem statement implemented (chapter 2)	Yes
Separate secure login for every role	Yes
Database scripts: EF migrations and Database\NexusServiceMarketingDb-schema.sql	Yes
Every code block commented	Yes
Unit testing checklist completed - all tests pass	Yes
Interface tested on phone, tablet and desktop widths	Yes
Project report, presentation, status report and feedback form included	Yes
Placeholders (name, enrollment no., batch, group, centre, dates) filled in	[]
Database backup (.bak) added to the submission folder	[]
Submission ZIP named Batch_Group_NexusServiceMarketingSystem.zip	[]

Appendix - Plan price list

Service	Plan	Price	Validity	Deposit
Dial-Up	Hourly 10 / 30 / 60 hrs	\$50 / \$130 / \$260	1 / 3 / 6 months	\$325
Dial-Up	Unlimited 28 Kbps monthly / quarterly	\$75 / \$150	1 / 3 months	\$325
Dial-Up	Unlimited 56 Kbps monthly / quarterly	\$100 / \$180	1 / 3 months	\$325
Broadband	Hourly 30 / 60 hrs	\$175 / \$315	1 / 6 months	\$500
Broadband	Unlimited 64 Kbps monthly / quarterly	\$225 / \$400	1 / 3 months	\$500
Broadband	Unlimited 128 Kbps monthly / quarterly	\$350 / \$445	1 / 3 months	\$500
Landline	Local: yearly \$75 (0.55/min), monthly \$35 (0.75/min)	\$75 / \$35	12 / 1 months	\$250
Landline	STD monthly \$125, half-yearly \$420 (rates per minute)	\$125 / \$420	1 / 6 months	\$250